

Open-Source USB-Programmable RF Signal Generator — 35 MHz to 4.4 GHz

The RFGEN44 is an open-source, USB-programmable RF signal generator capable of generating frequencies up to 4.4 GHz. It gives anyone working with RF a quick, easy signal source and operates completely driver-free over USB HID — no drivers or special permissions on any generic OS. It is fully programmable under Windows, Linux, and macOS via the provided Python scripts or the cross-platform Qt5 GUI (built on libHID), and raw HID support enables control from any scientific platform, including MATLAB. The hardware is based on the Analog Devices ADF4351 wideband synthesizer, driven by a USB microcontroller with high-endurance flash for standalone operation.

KEY FEATURES

- ADF4351 wideband synthesizer with integrated VCO
- USB MCU with high-endurance flash memory
- 35 MHz to 4.4 GHz output, 10 kHz step
- Complementary RF outputs (RF Out +/–, 180° apart)
- Adjustable output amplitude in 4 steps
- AUX (Ref/Trigger) — 4 functions: ext reference in (10–250 MHz), int reference out, sweep/macro sync out, sync in
- Automatic frequency sweep & hop, software or standalone
- On-device programmable macros: step sequences (freq, hold time, RF on/off), standalone or PC
- Standalone operation, auto-config from flash
- Driver-free USB HID, no drivers on any generic OS
- Cross-platform Qt GUI + Python + raw HID (MATLAB)
- Multiple devices per PC with identification LED
- Open-source hardware and software



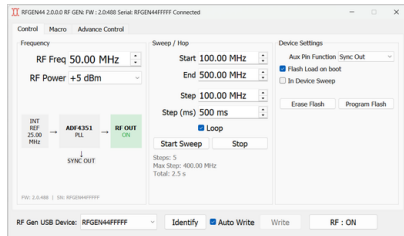
RFGEN44 — compact aluminium enclosure, 92 × 48 × 26 mm

SPECIFICATIONS

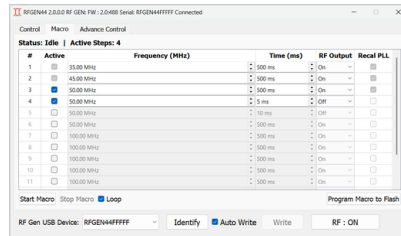
Synthesizer	Analog Devices ADF4351, wideband, integrated VCO
Controller	USB MCU, high-endurance flash
Frequency Range	35 MHz to 4.4 GHz
Frequency Step	10 kHz
RF Outputs	Two complementary outputs (RF Out + / RF Out –), 180° apart
Output Amplitude	Adjustable, 4 steps
Max Output Power	35 MHz–1 GHz: –1 dBm 1–2 GHz: 0.0 dBm 2–4.4 GHz: –4.0 dBm
Reference Clock	Internal 25 MHz ±50 ppm oscillator, or external high-stability timebase via AUX — highly flexible external reference input range of 10 MHz to 250 MHz; internal timebase can also be output
Sweep / Hop	Automatic frequency sweep and hop; software or standalone
Programmable Macros	User-programmed on-device step sequences — each step sets frequency, hold time, and RF output on/off; executes single-shot, looped, or trigger-advanced (via Ref/Trigger), with or without a PC
AUX (Ref/Trigger)	SMA female, TTL 3.3 V. Four functions: external reference in (10–250 MHz), internal reference out, sweep/macro sync out, and sync in
Standalone Mode	Auto-config load from flash; no PC required
Host Interface	USB HID (driver-free)
USB Connector	USB Type-C
RF Connectors	SMA female (RF Out +, RF Out –)
Power Supply	5 V, USB bus-powered
Indicators	Power LED; Status / identification LED
Dimensions	92 × 48 × 26 mm
Weight	65 g
Compliance	CE / UKCA marked; RoHS; WEEE registered
Country of Origin	Germany

FIRMWARE & SOFTWARE

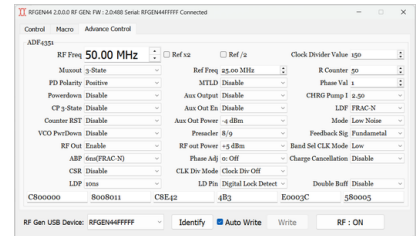
Firmware on the USB MCU runs a USB stack that bridges USB to SPI to control the ADF4351 PLL, with high-endurance flash for standalone operation. The AUX (Ref/Trigger) pin is software-selectable between four functions: external reference input, which accepts a high-stability timebase over a flexible 10–250 MHz range as an alternative to the on-board 25 MHz ± 50 ppm oscillator; internal reference output, to discipline other instruments from the RFGEN44; sweep/macro sync output, to trigger external equipment such as an oscilloscope or spectrum analyzer in step with the device; and sync input, to advance the device from an external trigger. Beyond simple sweep and hop, the firmware includes a programmable macro engine: the user defines a sequence of steps — each specifying frequency, hold time, and RF output on/off — which the device executes single-shot, looped, or advanced by an external trigger, entirely on its own or under PC control. The open-source Qt application programs frequency in 10 kHz steps, configures sweep and hop with programmable dwell and step, selects the AUX function, and writes the standalone configuration to the device. Full Qt and Python source ship with pre-built binaries for control from any HID-capable environment, including MATLAB.



Control — frequency, power, sweep/hop

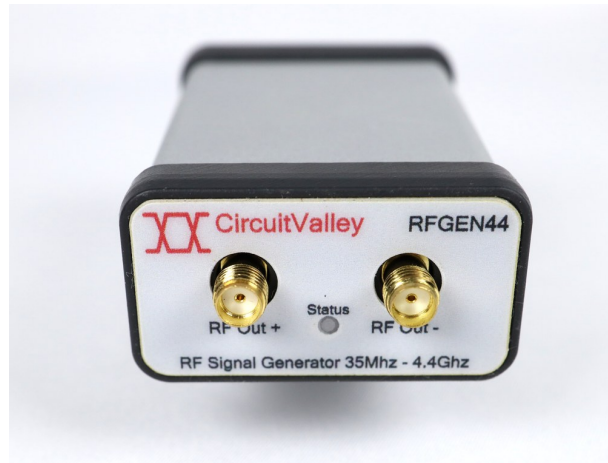


Macro — programmable step sequences

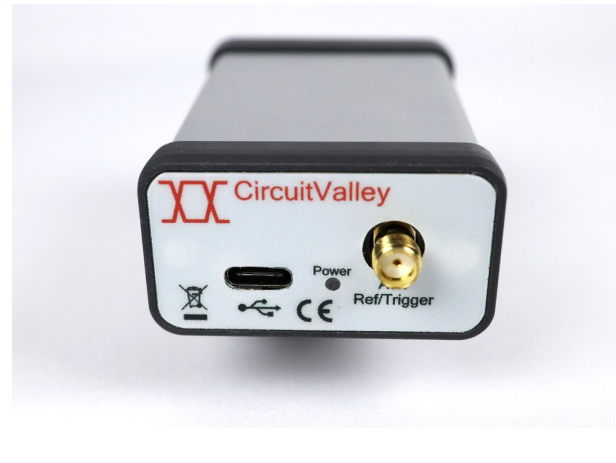


Advance Control — full ADF4351 registers

CONNECTIONS & INTERFACES



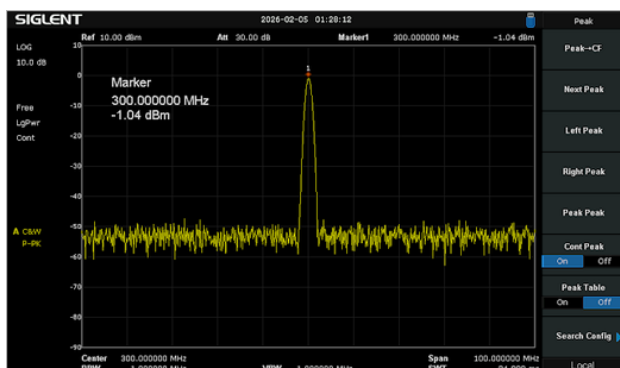
Front panel. Two complementary RF outputs on SMA female connectors: **RF Out +** and **RF Out -**, 180° out of phase. Centre **Status** LED indicates device and output state.



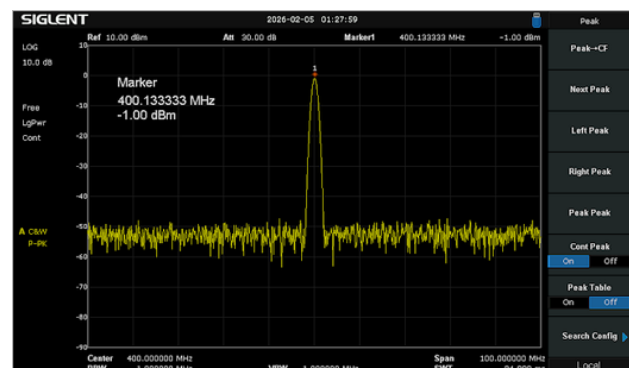
Rear panel. **USB Type-C** for power and driver-free HID control, **Power** LED, and the **Ref/Trigger** port (SMA female, TTL 3.3 V) for its four functions: external reference in, internal reference out, and sweep/macro sync out and in. CE and WEEE marked.

MEASURED RF OUTPUT

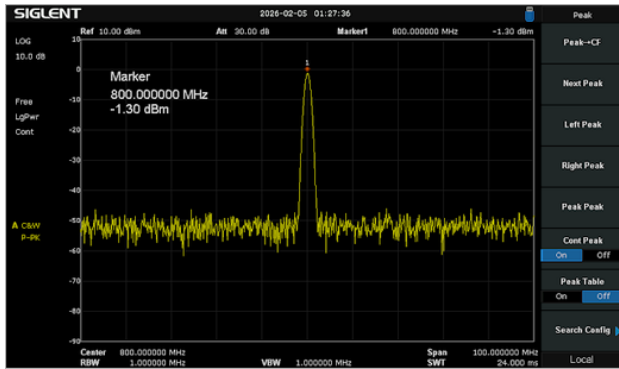
Single-tone output measured on a Siglent SSA3032X spectrum analyzer across the band:



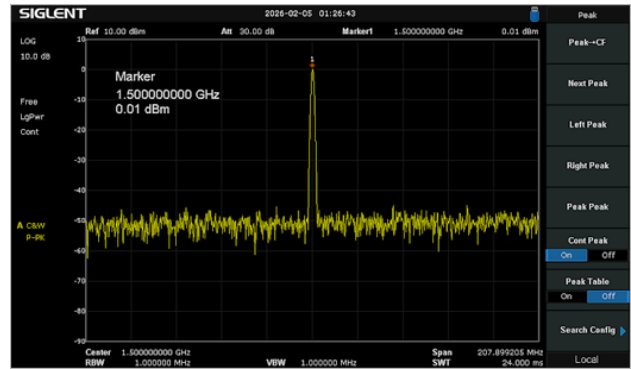
300 MHz — -1.04 dBm



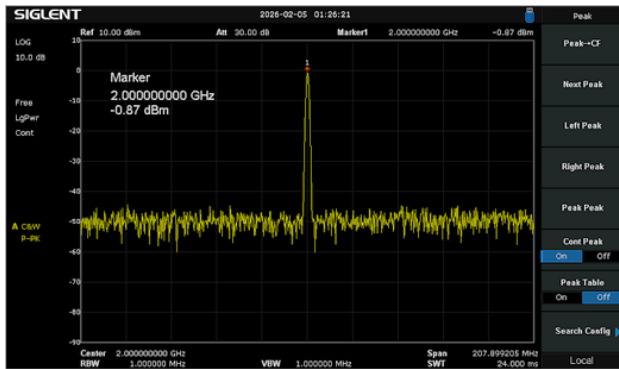
400 MHz — -1.00 dBm



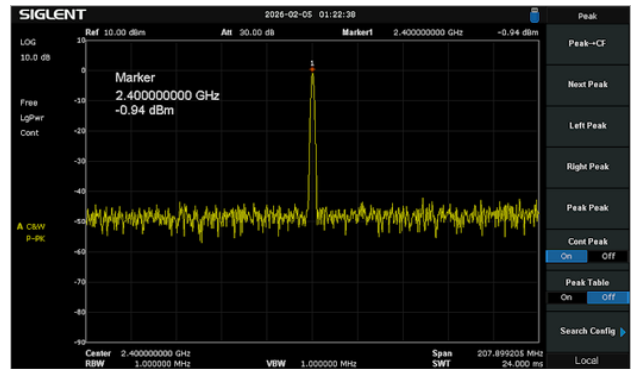
800 MHz — -1.30 dBm



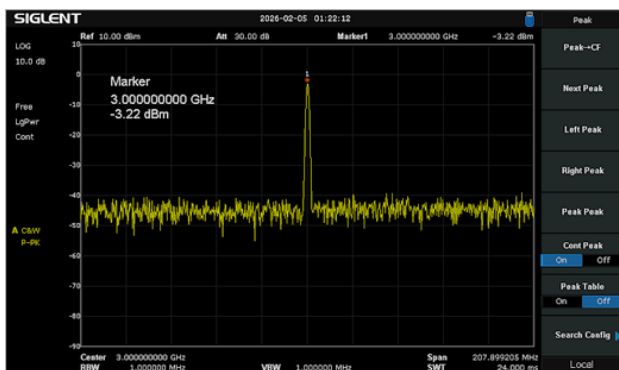
1.5 GHz — +0.01 dBm



2.0 GHz — -0.87 dBm



2.4 GHz — -0.94 dBm



3.0 GHz — -3.22 dBm

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